

PEIYUAN LEE

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES

Los Angeles, CA

Bachelor of Science, Statistics & Data Science, Minor in Data Science Engineering

June 2025

- **GPA/ Major GPA:** 3.75/ 3.89
- **Awards:** Dean's Honors List, ASA DataFest Finalist
- **Coursework:** Neural Networks, Machine Learning, Linear Algebra, Data Mining, Experimental Design and Analysis, Database Systems, Algorithms and Complexities, Probability Theory, Bayesian Statistics, Causal Inference

EXPERIENCE

Data Engineering Assistant

Los Angeles, CA

Backus Lab at UCLA

Jun 2023 – Dec 2024

- Developed scalable ETL pipeline to integrate large, complex protein datasets, leveraging Python and SQL for predictive analysis
- Built a full-stack web application (Django, JavaScript, HTML/CSS) enabling real-time lookup and merging of 5,000+ entries
- Containerized the entire system using Docker to streamline deployment and ensure consistent reproducibility across the team
- Co-authored paper accepted by *Nature Computational Science*, applying machine learning to predict protein behavior ([link](#))

Data Science/Software Engineering Intern

Kuala Lumpur, Malaysia

Pantas Softwares

Jul 2023 – Aug 2023

- Developed and integrated OCR feature into the company's website using Django, AWS, and JavaScript to enable automated text extraction from uploaded financial documents
- Fine-tuned OCR extraction models to achieve ~98% accuracy, enhancing data processing efficiency and user experience
- Collaborated with a cross-functional team of 10+ team members to review code and deploy production-ready features

NLP Research Assistant

Los Angeles, CA

Amazon X UCLA

Nov 2022 – Mar 2023

- Analyzed 200+ short stories to identify narrative features (e.g., emotional arc, themes) correlated with story engagement
- Designed end-to-end NLP pipelines with BERT, GPT-3, and Gensim to extract thematic and emotional trends in short stories
- Consolidated model outputs into research insights, presented visualizations and results directly to lead Amazon researcher

Data Analytics Intern

Kuala Lumpur, Malaysia

Securities Commission Malaysia

Jun 2022 – Aug 2022

- Scraped and cleaned unstructured textual data for supervised machine learning models using pandas, SQL, and BeautifulSoup
- Built automated ETL pipeline in Python to extract corporate disclosures, reducing manual labeling time by 80%
- Produced literature review for text extraction algorithms and presented insights to engineering and data compliance teams

PROJECTS

Electroencephalogram (EEG) Signal Classification

- Built a hybrid CRNN architecture using TensorFlow to classify EEG signals, achieving accuracy of ~ 72.14% across 4 classes
- Increased dataset size using data augmentation techniques and Variational Autoencoders (VAEs) to enhance model generalization
- Enhanced model performance by ~10% through hyperparameter tuning with KerasTuner, improving reliability of predictions

Spotify Genre Predictions

- Performed EDA, preprocessed data and built data visualizations on Spotify data with 114,000+ rows and 20 features
- Addressed class imbalance through SMOTE and custom probability thresholding to boost model fairness and precision
- Benchmarked algorithms like random forests, clustering analysis and PCA to achieve accuracy of ~70% across 20 classes

Motility Restriction in Orbital Fracture Patients Analysis

- Spearheaded analysis of CT imaging data from 123 patients with orbital fractures to assess association with eye motility restriction
- Built 3 machine learning models to predict presence of motility restriction; identified 4+ key predictors using SHAP analysis
- Conducted statistical tests (t-tests, ANOVA) to validate findings; visualized results with Python to aid non-technical interpretation
- Synthesized insights into a clinical report for presentation to collaborating ophthalmologists and research stakeholders

Financial Valuation Modeling

- Trained and fine-tuned XGBoost, Random Forests, and linear regression models to predict company valuations to better inform investor decision-making, achieving an average R^2 score of 98% across models
- Built valuation prediction pipeline using pandas to process venture funding data with 30,000+ rows and 60 financial features
- Conducted hypothesis testing (t-tests, ANOVA) and visualized valuation trends to uncover patterns across the different features

SKILLS

- **Languages:** Python, C++, R, SQL, HTML/CSS, JavaScript
- **Framework:** Tensorflow, Keras, Django, NumPy, Pandas, Matplotlib, PyTorch, Bootstrap, JQuery, Selenium, Scikit-learn
- **Tools:** Git/Github, Jira, Docker, AWS services, Microsoft Excel, Word, Powerpoint, Access, Tableau, PowerBI